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Midterm

1.

Recall that if $f(\mathbf{x})$ and $g(\mathbf{x})$ are probability distributions, then the Kullback-Leibler divergence of g from f is

$$D(f,g) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(\mathbf{x}) \log[\frac{f(\mathbf{x})}{g(\mathbf{x})}] d\mathbf{x}.$$

We seek to approximate an arbitrary distribution $f(\mathbf{x})$ by a normal distribution $g(\mathbf{x}) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Show that the values that minimize $D(f,g)$ are

$$\boldsymbol{\mu} = \mathbb{E}_{f(\mathbf{x})}[\mathbf{x}],$$

$$\boldsymbol{\Sigma} = \mathbb{E}_{f(\mathbf{x})}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T]$$

Given $D(f,g)$, let's first try to simplify so it's less intimidating to understand:

$$D(f,g) = \int_{-\infty}^{\infty} f(\mathbf{x}) \log[\frac{f(\mathbf{x})}{g(\mathbf{x})}] d\mathbf{x}$$

$$= \int_{-\infty}^{\infty} f(\mathbf{x}) \log(f(\mathbf{x})) d\mathbf{x} - \int_{-\infty}^{\infty} f(\mathbf{x}) \log(g(\mathbf{x})) d\mathbf{x}$$

From here, we aim to minimize D with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, which are both parameters of $g(\mathbf{x})$, so we note that we only really need to care about optimizing in terms of $g(\mathbf{x})$. Looking back at our above terms, we note that this is all in terms of the arbitrary $f(\mathbf{x})$, so there's nothing to be done here in terms of $g(\mathbf{x})$. We are subtracting the second term, so minimizing $D(f,g)$ overall means maximizing

$$\int_{-\infty}^{\infty} f(\mathbf{x}) \log(g(\mathbf{x})) d\mathbf{x}$$

which as hinted at by the expectations in terms of $f(\mathbf{x})$ can be translated into

$$= \mathbb{E}_{f(\mathbf{x})}[\log(g(\mathbf{x}))]$$

We substitute in the definition for the multivariate normal distribution for $g(\mathbf{x})$, as taken from 2.5.2: Multivariate Density in Pattern Classification, by Duda, et al.:

$$= \mathbb{E}_{f(\mathbf{x})}[\log(\frac{1}{(2\pi)^{k/2}|\boldsymbol{\Sigma}|^{1/2}} \exp(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T)\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}))]$$

2.

The columns in the file `midterm_exam_data.txt` are the X and Y values of a function which is a sum of cosines with added Gaussian noise.

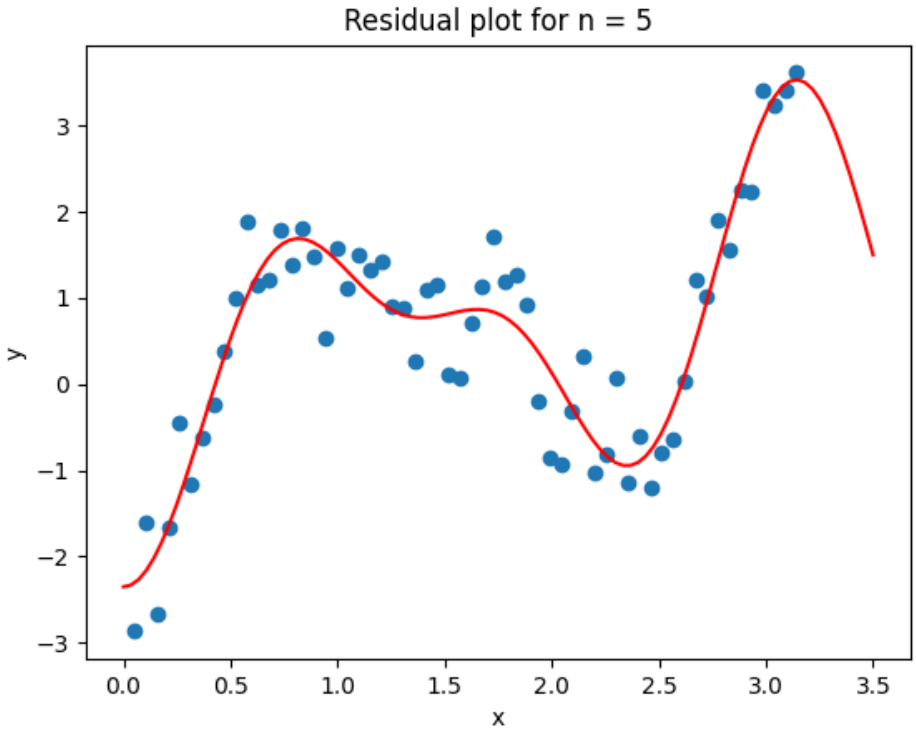
(a)

For various values of n, fit an nth order cosine series ($y = \frac{a_0}{2} + \sum_{k=1}^{n} a_k \cos kx$) to the data. What would you say is the most likely value of n?

I ran a python script and got the following output:

```
> python problem_2.py
          rss      rse  adj_r_2
n
0   115.87160  1.40140  0.00000
1   107.03410  1.35846  0.06034
2   106.96723  1.36990  0.04446
3    38.57145  0.82993  0.64929
4    37.30363  0.82356  0.65465
5    13.40291  0.49820  0.87362
6    13.23616  0.49974  0.87284
7    13.18306  0.50351  0.87091
8    13.10839  0.50698  0.86913
9    12.84362  0.50683  0.86920
10   12.78047  0.51071  0.86719
```

I chose to do this by looking at the measures used in section 3.1.3: Assessing the Accuracy of the Model from James, G. et. al, An Introduction to Statistical Learning: with Applications in R, Springer. Looking at this, the RSS seems to really improve and stabilize around $\boxed{n = 5}$ and not really improve drastically afterwards. This is the same case with the R squared and the RSE values too. I also plotted to see if this seemed sane, and it definitely seems to fit pretty well: I got this from



running the following python code:

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import statsmodels.api as sm

data = np.loadtxt("midterm_exam_data.txt")

x_data = data[:, 0]
y_data = data[:, 1]

def cos_function(x_data, n):
  """BUild the corresponding matrix by stacking 1, cos(x), etc."""
  return np.column_stack(
    [np.ones_like(x_data)] +
    [np.cos(k * x_data) for k in range(1, n + 1)]
```

```
  )

# Try for n-values up to 10
fit_results = []
result_summaries = []
for n in range(0, 11):
  fit_result = sm.OLS(y_data, cos_function(x_data, n)).fit()
  fit_results.append(fit_result)

  result_summaries.append({
    "n": n,
    "rss": fit_result.ssr,
    "rse": np.sqrt(fit_result.scale),
    "adj_r_2": fit_result.rsquared_adj,
  })

result_summary_df = pd.DataFrame(result_summaries).set_index("n").round(5)
print(result_summary_df)

# n = 5 seems reasonable, so plot and see
plt.scatter(x_data, y_data)
x_series = np.linspace(0, 3.5, 100)
plt.plot(x_series, fit_results[5].predict(cos_function(x_series, 5)), color=
plt.xlabel("x")
plt.ylabel("y")
plt.title("Residual plot for n = 5")
plt.show()
```

(b)

Estimate the level of the noise.

The output from my script was as follows:

```
Estimated noise standard deviation: 0.49819887898041026
Estimated noise variance: 0.24820212301733746
```

from the following block of code at the bottom of my python script:

```
# Print RSE as noise estimation
sigma = np.sqrt(fit_results[5].scale)
print(f"Estimated noise standard deviation: {sigma}")
print(f"Estimated noise variance: {sigma**2}")
```

The noise seems relatively low in terms of standard deviation. In addition, looking back at our table in part (a) with the RSE's seeming also relatively stable from $n = 5$ onwards, which also seems to indicate that the noise is relatively low. I chose to use RSE as a measure of noise used in section 3.1.3: Assessing the Accuracy of the Model from James, G. et. al, An Introduction to Statistical Learning: with Applications in R, Springer

3.

Consider a Bayesian classifier, with distributions being n-dimensional multivariate Bernoulli random variables. The conditional probability for a given category is

$$P(\mathbf{x}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{x_i}(1-\theta_i)^{1-x_i},$$

and let $\mathcal{D} = \{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ be a set of k samples independently drawn from this probability density.

(a)

If $\mathbf{s} = (s_1, \ldots, s_n)^T$ is the sum of the k samples, show that

$$P(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i},$$

From the "Review of Statistics" lecture from Module 2, we know the following: With the setup of the problem with independent samples drawn from a Bernoulli distribution, this is simply the likelihood function. We can therefore just express this as the product of the probability densities:

$$P(\mathcal{D}|\boldsymbol{\theta}) = \mathcal{L}(\boldsymbol{\theta}|D) = \prod_{j=1}^{k} P(\mathbf{x}_j|\boldsymbol{\theta})$$

$$= \prod_{j=1}^{k}(\prod_{i=1}^{n} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}) = \prod_{j=1}^{k}\prod_{i=1}^{n} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}$$

$$= \prod_{i=1}^{n}\prod_{j=1}^{k} \theta_i^{x_{ij}}(1-\theta_i)^{1-x_{ij}}$$

$$= \prod_{i=1}^{n} \theta_i^{\sum_{j=1}^{k} x_{ij}}(1-\theta_i)^{\sum_{j=1}^{k} 1-x_{ij}}$$

Substituting the sum s in accordingly as given gives us exactly the desired result:

$$\boxed{P(\mathcal{D}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}$$

(b)

Assuming a uniform prior distribution for $\boldsymbol{\theta}$ show that

$$P(\boldsymbol{\theta}|\mathcal{D}) = \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i},$$

From part (a), we can find this using Bayes' thoerem (Chapter 2: Bayesian Decision Theory from Pattern Classification, by Duda, et al.):

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}{P(\mathcal{D})}$$

$$= \frac{P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}{\int P(\mathcal{D}|\boldsymbol{\theta}) \cdot P(\boldsymbol{\theta})}$$

Substituting in our result from part (a):

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot P(\boldsymbol{\theta})}{\int \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot P(\boldsymbol{\theta})}$$

And using the assumption of the uniform distribution for $\boldsymbol{\theta}$, i.e. $P(\boldsymbol{\theta}) = 1$ for $0 \leq \theta_i \leq 1$:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot 1}{\int \prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i} \cdot 1}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \int_0^1 \theta_i^{s_i}(1-\theta_i)^{k-s_i} d\theta_i}$$

We use integration by parts in the denominator:

- $u = (1-\theta_i)^{k-s_i}$
- $du = (1-\theta_i)^{k-s_i-1} \cdot (k-s_i) \cdot -1 d\theta_i = (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i$
- $v = \frac{\theta_i^{s_i+1}}{s_i+1}$
- $dv = \theta_i^{si} d\theta_i$ And substituting accordingly:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n}(1-\theta_i)^{k-s_i} \cdot \frac{\theta_i^{s_i+1}}{s_i+1}|_0^1 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} 0 - 0 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} 0 - 0 - \int_0^1 \frac{\theta_i^{s_i+1}}{s_i+1} \cdot (s_i-k)(1-\theta_i)^{k-s_i-1} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} -\frac{s_i-k}{s_i+1} \int_0^1 \theta_i^{s_i+1}(1-\theta_i)^{k-s_i-1} d\theta_i}$$

We can see that from repeatedly applying integration by parts, we eventually reduce the exponent of the $1-\theta_i$ term to 0, while the exponent of the θ_i term increases until k, giving us:

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)\ldots 1}{(s_i+1)(s_i+2)\ldots k} \int_0^1 \theta_i^{k}(1-\theta_i)^{0} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)\ldots 1}{(s_i+1)(s_i+2)\ldots k} \int_0^1 \theta_i^{k}(1-\theta_i)^{0} d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} \int_0^1 \theta_i^k d\theta_i}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} [\frac{\theta_i^{k+1}}{k+1}]_0^1}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)(k-s_i-1)...1}{(s_i+1)(s_i+2)...k} (\frac{1}{k+1} - 0)}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!}{(s_i+1)(s_i+2)...k+1}}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!}{(s_i+1)(s_i+2)...k+1} \cdot \frac{s_i!}{s_i!}}$$

$$= \frac{\prod_{i=1}^{n} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\prod_{i=1}^{n} \frac{(k-s_i)!s_i!}{(k+1)!}}$$

$$= \prod_{i=1}^{n} \frac{\theta_i^{s_i}(1-\theta_i)^{k-s_i}}{\frac{(k-s_i)!s_i!}{(k+1)!}}$$

$$\boxed{P(\boldsymbol{\theta}|\mathcal{D}) = \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \theta_i^{s_i}(1-\theta_i)^{k-s_i}}$$

exactly as we sought to find.

(c)

Plot this density for the case $n = 1, k = 1$ and for the two resulting possibilities for s_i.

With $n = 1$, we have one sample, so our only cases are either $s = 0$ or $s = 1$:

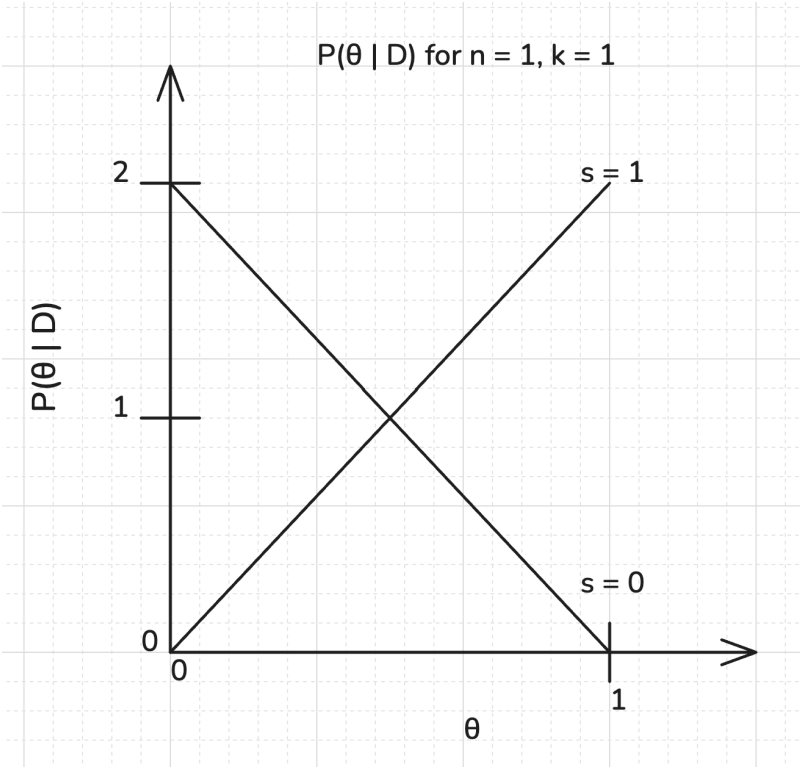
- If $s = 0$, we have

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{(1+1)!}{0!(1-0)!} \cdot \theta^0 \cdot (1-\theta)^{1-0} = \frac{2!}{0!1!} \cdot 1 \cdot (1-\theta)^1 = 2 - 2\theta$$

- If $s = 1$, we have

$$P(\boldsymbol{\theta}|\mathcal{D}) = \frac{(1+1)!}{1!(1-1)!} \cdot \theta^1 \cdot (1-\theta)^{1-1} = \frac{2!}{1!0!} \cdot \theta \cdot (1-\theta)^0 = 2\theta$$

We plot this below:



(d)

Integrate the product $P(\mathbf{x}|\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathcal{D})$ over $\boldsymbol{\theta}$ to obtain

$$P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n}(\frac{s_i+1}{k+2})^{x_i}(1-\frac{s_i+1}{k+2})^{1-x_i}$$

As directed, we start with

$$P(\mathbf{x}|\mathcal{D}) = \int_{\boldsymbol{\theta}} P(\mathbf{x}|\boldsymbol{\theta})P(\boldsymbol{\theta}|\mathcal{D})d\boldsymbol{\theta}$$

(2.10: Missing and Noisy Features from Pattern Classification, by Duda, et al.) Substituting the given conditional probability and our finding from part (b):

$$= \int_{\boldsymbol{\theta}} (\prod_{i=1}^{n} \theta_i^{x_i}(1-\theta_i)^{1-x_i}) \cdot (\prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i})d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \int_{\boldsymbol{\theta}} \theta_i^{x_i}(1-\theta_i)^{1-x_i} \frac{(k+1)!}{s_i!(k-s_i)!}\theta_i^{s_i}(1-\theta_i)^{k-s_i}d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \int_{\boldsymbol{\theta}} \theta_i^{x_i}(1-\theta_i)^{1-x_i}\theta_i^{s_i}(1-\theta_i)^{k-s_i}d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \int_{\boldsymbol{\theta}} \theta_i^{x_i+s_i}(1-\theta_i)^{k-s_i-x_i+1} d\boldsymbol{\theta}$$

This leads us into a very similar situation as we saw in part (b) with the repeated integration by parts. In part (b), we have exponents of s_i and $k-s_i$; here, we have exponents of s_i+x_i and $k-s_i-x_i+1$, so the $1-\theta_i$ term now has 1 additional power relative to the θ_i term as it did in part (b). We just integrate by parts $k-s_i-x_i+1$ times, giving us

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+1)} \int_0^1 \theta_i^{k+1}(1-\theta_i)^0 d\boldsymbol{\theta}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+1)} [\frac{\theta_i^{k+2}}{k+2}]_0^1$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+1)} \cdot (\frac{1}{k+2} - 0)$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+2)}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!}{(s_i+x_i+1)(x_i+s_i+2)\dots(k+2)} \cdot \frac{(s_i+x_i)!}{(s_i+x_i)!}$$

$$= \prod_{i=1}^{n} \frac{(k+1)!}{s_i!(k-s_i)!} \cdot \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{(k+2)!}$$

$$= \prod_{i=1}^{n} \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{s_i!(k-s_i)!(k+2)}$$

$$= \prod_{i=1}^{n} \frac{(k-s_i-x_i+1)!(s_i+x_i)!}{s_i!(k-s_i)!(k+2)}$$

From this point, we can take a hint from the structure of the answer to see that we should be looking at the two different cases for x_i, that is, because it is Bernoulli, it must either be 1 or 0. Looking at these cases:

- If $x_i = 0$, we get

$$\frac{(k-s_i-0+1)!(s_i+0)!}{s_i!(k-s_i)!(k+2)} = \frac{(k-s_i+1)!s_i!}{s_i!(k-s_i)!(k+2)} = \frac{k-s_i+1}{k+2}$$

- If $x_i = 1$, we get

$$\frac{(k - s_i - 1 + 1)!(s_i + 1)!}{s_i!(k - s_i)!(k + 2)} = \frac{s_i + 1}{k + 2}$$

And we then can express this using the exponents to account for the 0 / 1 on/off nature of the results. We therefore get

$$= \prod_{i=1}^{n} (\frac{k - s_i + 1}{k + 2})^{1-x_i} (\frac{s_i + 1}{k + 2})^{x_i}$$

$$= \prod_{i=1}^{n} (\frac{s_i + 1}{k + 2})^{x_i} (\frac{k - s_i + 1}{k + 2})^{1-x_i}$$

$$= \prod_{i=1}^{n} (\frac{s_i + 1}{k + 2})^{x_i} (\frac{k + 2 - s_i - 1}{k + 2})^{1-x_i}$$

$$\boxed{P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n} (\frac{s_i + 1}{k + 2})^{x_i} (1 - \frac{s_i + 1}{k + 2})^{1-x_i}}$$

exactly as we sought to find.

(e)

Compare the expressions for $P(\mathbf{x}|\mathcal{D})$ and $P(\mathbf{x}|\boldsymbol{\theta})$. What is $\hat{\boldsymbol{\theta}}$, the effective Bayesian estimate for $\boldsymbol{\theta}$?

To recap, we have

$$P(\mathbf{x}|\mathcal{D}) = \prod_{i=1}^{n} (\frac{s_i + 1}{k + 2})^{x_i} (1 - \frac{s_i + 1}{k + 2})^{1-x_i}$$

and

$$P(\mathbf{x}|\boldsymbol{\theta}) = \prod_{i=1}^{n} \theta_i^{x_i} (1 - \theta_i)^{1-x_i}$$

We see that that θ_i directly maps to $\frac{s_i+1}{k+2}$ for both the θ_i and $1 - \theta_i$ terms, which indicates to us that $\hat{\boldsymbol{\theta}}$ is $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n)^T$ where $\hat{\theta}_i = \frac{s_i+1}{k+2}$.